



Skills

</> Languages

C# C C++ .NET ShaderCore HLSL Rust Java
JavaScript HTML CSS PHP Python

Software

Unity Unreal Engine 4 Greenfoot Git Flow Jenkins Jira
Game Engines Version Control
Visual Studio VSC MySQL React ASP.NET Blazor
IDEs & Frameworks
Photoshop Cinema 4D Blender FileZilla Audacity OBS
Art Media
Premier Pro Sony Vegas Davinci Resolve 16
Editing

Professional
Advanced
Intermediate
Beginner

Experience



Professional Projects

Game Developer & Software Engineer

Red Cartel

August 2024 - March 2025



Red Cartel, xReality Group - NSW, Australia

- Worked in a team of 20+ including programmers, designers, artists, engineers, and admin.
- Collaborated with clients to design and implement core gameplay features and resolve bugs.
- Minimized user blunder by providing operators with tools for monitoring user training progress.
- Lead the development of automated gameplay unit testing to eliminate redundancies within development and quality assurance.
- Reconstructed multi-platform camera system into a modular perspective system that greatly improved operator accessibility and user experience.
- Maintained Asset Bundles used for cross platform data sharing, allowing for optimised performance and reduced customer frustration.

Game Developer

Powercor

April 2023 - July 2024



Varsity
plazr

Powercor Australia - Victoria, Australia

- Redesigned and implemented core gameplay loop to better improve the user's interactive experience.
- Improved data collection and visibility to allow for better user feedback and results.
- Prevented user data loss by caching data locally to improve user retention and reduce frustration.
- Dramatically improved the user interface and load times for the customer when reviewing user results.
- Implemented unit testing for Git Flow validation to maintain a healthy production environment for the customer.
- Refactored legacy codebase to improve maintainability and remove code debt.
- Documented design processes and implementation of developed and legacy code to improve future productivity.

Jr Game Developer

Powercor

Feburary 2023 - April 2023

- Worked closely with a dedicated team of designers, lead programmer, and project manager.
- Lead a small team of designers to redesign and implement an improved user interface to better the user experience.
- Designed and Implemented multi-platform tool interactions aimed to improve accessibility and visual clarity for the user.
- Updated and refactored VR experience to improve user interaction and features.
- Onboarded and trained multiple developers including a senior.

ScriptableObject Editor

Indie

July 2022 - December 2022



An object management editor tool published on the Unity Asset Store.

- Designed to simplify Scriptable Object modification to improve developer workflow.
- Aid game designers with folder management, object filtering, and type masking.
- Multi-window user interface for centralized control over scriptable objects.
- Supports user created assets and third party / custom packages.
- Tested and built for all versions of Unity past '2019.4'.
- Provides the user with settings to tailor the experience towards the developer's needs.
- Designed and Implemented modularly allowing for user driven additions to be made.
- Easily customisable UI designed to be bent to the user's visual taste.



University Projects

Summit - Lead Programmer

MQU

July 2022 - November 2022



Student project with 5 team members, 3 programmers. 3D third person rogue-like made in Unity. 410+ personal hours dedicated.

- Re-engineered existing codebase focusing on efficiency, modularity, and simplicity.
- Acted as senior developer to manage & aid development processes across the board (including design and art).
- Designed and implemented all backend systems related to gameplay and structure.
- Designed and implemented custom Boss State Machine.
- Modularized boss states for designer friendly implementation.
- Implemented modular animation event listeners to control states
- Implemented modularized scene management with asynchronous loading.
- Designed and assisted implementation of power-ups.
- Designed and implemented a custom interaction system utilizing listeners for modularity.
- Developed announcement and dialogue systems.
- Custom camera handler for panning, effects, and 3rd person perspective.
- Managed custom lighting and skybox handling.

CQU Statistics Project - Lead Programmer

MQU

Feburary 2022 - June 2022



Student project with 5 team members, 1 programmers. Gamification visual novel made in Unity.

- Apart of the PACE Internship program at Macquarie University, dedicated client was Central Queensland University.
- Designed, implemented, and maintained systems in accordance to requirements given by the client.
- Client task: Create a gamified quiz to track student's knowledge of statistics.
- Custom Dynamic Dialogue System with dialogue.
- Full intergration with Google Sheets API v4.
- Records and collects player completion data using google database.
- Completely automated visual novel setup without additional programming.
- Dialogue options used to create diverging pathways for player progress based on their performance.
- Custom commands and modifiers used to enhance dialogue with effects, sprite changes, and more.
- Framework creates game content based on .json level files.
- Full intergration with Json.NET framework using custom class & interface based serialization.



Personal Projects

Dark Light

July 2019 - Present

2D metroidvania inspired by Hollow Knight and the Soulsbourne Series. Paused development while at Univeristy.

- Custom player state machine.
- Custom velocity based movement for fast paced mobility and combat.
- Command based custom input system used to handle player movement. Allows for control over the player to be utilized in systems handling cutscenes and event triggers.
- Custom input system derived from Unity Events to allow for control configuration and keybinding storing.
- Modular interface based save system using Json.NET (Works with binary serialization).
- From scratch A* pathfinding for non-player entities.

Education

Bachelor of Game Design & Development

2020 - 2022

Macquarie University

Sydney, Australia

Diploma of Gaming Development

2019

Northern Sydney Institute TAFE

Sydney, Australia